**Parul Institute of Engineering & Technology**

**Department of Computer Science & Engineering**

**Design & Analysis of Algorithm Laboratory**

**(303105219)**

## Practical Observation Table

| Sr. No | Experiment Title | Page No | | Date of Start | Date of Completion | Sign | Marks (out of 10) |
|---|---|---|---|---|---|---|---|
| | | From | To | | | | |
| 1. | Write a program to determine whether the given number is prime or not | | | | | | |
| 2. | Given a sorted array and a target value, return the index if the target is found. If not, return the index where it would be if it were inserted in order | | | | | | |
| 3. | There are N children standing in a line with some rating value. You want to distribute a minimum number of candies to these children such that: Each child must have at least one candy. The children with higher ratings will have more candies than their neighbours. You need to write a program to calculate the minimum candies you must give. | | | | | | |

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| **4.** | There is a new barn with N stalls and C cows. The stalls are located on a straight line at positions x1,xN(0 <= xi <= 1,000,000,000). We want to assign the cows to the stalls, such that the minimum distance between any two of them is as large as possible. What is the largest minimum distance? | | | | | | |
| **5.** | Given an undirected graph with V vertices and E edges, check whether it contains any cycle or not | | | | | | |
| **6.** | There are n servers numbered from 0 to n – 1 connected by undirected server-to-server connections forming a network where connections[ i ] = [ai, bi] represents a connection between servers ai and bi. Any server can reach other servers directly or indirectly through the network. A critical connection is a connection that, if removed, will make some servers unable to reach some other servers. Return all critical connections in the network in any order. | | | | | | |
| **7.** | Given a grid of size N x M (N is the number of rows and M is the number of columns in the grid) consisting of ‘0’s (Water) and ‘1’s(Land). Find the number of islands. | | | | | | |

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| **8.** | Given a grid of dimension N x M where each cell in the grid can have values 0, 1, or 2 which has the following meaning:<br>0: Empty cell<br>1: Cells have fresh oranges<br>2: Cells have rotten oranges<br>We have to determine what is the minimum time required to rot all oranges. A rotten orange at index [i, j] cannot other fresh oranges at indexes [i-1,j], [i+1,j], [i,j-1], [i,j+1] (up, down, left and right) in unit time | | | | | | | |
| **9.** | Given two strings str1 and str2 and below operations that can be performed on str1. Find minimum number of edits (operations) required to convert 'str1' into 'str2'. Insert Remove Replace, All of the above operations are of equal cost. | | | | | | | |
| **10.** | Minimum Path Sum" says that given a n x m grid consisting of non-negative integers and we need to find a path from top-left to bottom right, which minimizes the sum of all numbers along the path. | | | | | | | |
| **11.** | Given string num representing a non-negative integer num, and an integer k, return the smallest possible integer after removing k digits from num. | | | | | | | |

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| **12.** | There is a robot on an m x n grid. The robot is initially located at the top-left corner (i.e., grid[0][0]).The robot tries to move to the bottom-right corner (i.e., grid[m - 1][n - 1]). The robot can only move either down or right at any point in time. Given the two integers m and n, return the number of possible unique paths that the robot can take to reach the bottom-right corner. | | | | | | |